

Total no. of Pages: .....

Roll no. ....

Degree: B.Tech (CSDA / CSIOT)

Semester: 4<sup>th</sup> semester

**MID-SEMESTER EXAMINATION, Feb. 2025**

Course Title: DIGITAL LOGIC DESIGN

Course Code: CBEPC03 / CIEPC03 (OLD SCHEME)

Max. Marks: 15

Duration: 1:30 Hours

**Note:** - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

| Q. No. | Question                                                                                                                      | Marks | CO   |
|--------|-------------------------------------------------------------------------------------------------------------------------------|-------|------|
| 1a     | Given decimal numbers $X = 93$ and $Y = 47$ .<br>Find $(X+Y)$ in Binary and Hexadecimal values.                               | 1.5   | 1    |
| 1b     | If $A = 245$ and $B = 837$ , find the BCD sum of A and B.                                                                     | 1.5   | 1    |
| 2a     | For Octal numbers $x = (254)_8$ and $y = (173)_8$ ,<br>evaluate $(x - y)$ using 2's complement method.                        | 1.5   | 1    |
| 2b     | Implement logic circuit, using minimum NAND gates for:<br>$F(W, X, Y, Z) = \sum_{min} (2, 3, 5, 10, 12, 13) + d(0, 4, 9, 11)$ | 1.5   | 1    |
| 3a     | Design a logic circuit to obtain Gray code output, for<br>3-bits binary input.                                                | 1.5   | 1, 2 |
| 3b     | Draw the logic circuit of a 4-bit full adder/subtractor.                                                                      | 1.5   | 2    |
| 4a     | $F(P, Q, R) = \sum_{min} (0, 1, 2, 4, 7)$<br>Implement using a single 4:1 Multiplexer.                                        | 1.5   | 2    |
| 4b     | Design a Full-subtractor using a 1:8 De-MUX.                                                                                  | 1.5   | 2    |
| 5a     | Design a logic circuit to detect invalid BCD inputs.                                                                          | 1.5   | 2    |
| 5b     | Design a CMOS circuit to implement:<br>$F(W, X) = W \cdot X + W' \cdot X'$                                                    | 1.5   | 2    |